

PANKAJ KUMAR JANGID

Current position: PhD Research Scholar, IISER Pune India
E-mail: pkjangid151330@gmail.com
URL: jangidpk.github.io
Github: [JangidPK](https://github.com/JangidPK)

SUMMARY

Ph.D. researcher specializing in theoretical chemistry and stochastic modeling of molecular systems. My work combines statistical mechanics, mathematical modeling, and computational methods to investigate catalytic reactions, protein dynamics, molecular transport, and non-equilibrium processes. I am expanding my research toward physics-informed machine learning and AI-driven methods for predictive modeling and chemical discovery.

RESEARCH INTEREST

Machine Learning in Chemistry, Catalysis, Stochastic Processes, Molecular Simulations, Statistical Mechanics, Computational Chemistry.

RESEARCH EXPERIENCE

- 2021–Present **Ph.D. Researcher**, Indian Institute of Science Education and Research (IISER) Pune
- Developed analytical and stochastic models to investigate catalysis, protein dynamics, molecular transport, and non-equilibrium processes.
 - Derived theoretical frameworks based on statistical mechanics, stochastic processes, and reaction kinetics.
 - Performed numerical simulations and scientific computing using C++ to validate theoretical predictions.
 - Collaborated with international researchers on theoretical chemistry projects.
- 2020–2021 **Master's Thesis Researcher**, Indian Institute of Science Education and Research (IISER), Mohali
- Developed numerical methods to study quantum resonance states using the Siegert Pseudostate method in FORTRAN.
 - Analyzed resonance tunneling through phase-space distributions.

EDUCATION

- 2021 - present PhD (Chemical Physics) at **Indian Institute of Science Education and Research (IISER) Pune**
Thesis: Stochastic Dynamics and Nonequilibrium Transport in Complex Biological and Catalytic Systems
- 2016 - 2021 BS-MS Dual Degree at **Indian Institute of Science Education and Research (IISER) Mohali**
Major in chemistry, minor in physical chemistry.

SKILLS

Programming	Python (NumPy, SciPy, pandas, Matplotlib, PyTorch), C++, MATLAB, Mathematica
Machine Learning	PyTorch, Regression, Classification, PCA, Support Vector Machines (SVM), Fundamentals of Deep Learning (CNNs, RNNs, VAEs)
Analytical & Computational Modeling	Stochastic Processes, Brownian Dynamics, Monte Carlo Simulation, Non-Markovian Dynamics, Statistical Inference
Scientific Computing	Numerical Optimization, Scientific Computing, High-Performance Computing (HPC)
Mathematics	Probability Theory, Statistics, Statistical Mechanics, Differential Equations

SCIENTIFIC TALKS AND POSTERS

- 2025 Delivered a scientific talk at *Chemsymphoria 2025*, IISER Pune, Pune, India.
- 2025 Poster presentation at *Inter IISER–NISER Chemistry Meet*, IISER Pune, India.
- 2024 Poster presentation at *Society of Physical Chemistry (SoPhyC) Conference*, Indian Institute of Technology Bombay, Mumbai, India.
- 2024 Poster presentation at *RSC Meet the Editors*, IISER Pune & CSIR–National Chemical Laboratory, Pune, India.
- 2024 Poster presentation at *International Conference on Fundamental and Advanced Research in Chemistry*, Indian Institute of Technology Mandi, India.
- 2023 Poster presentation at *Theoretical Chemistry Symposium*, Indian Institute of Technology Madras, Chennai, India.

PUBLICATIONS

Research Highlights: 9 peer-reviewed publications (8 first-author); 2 additional first-author manuscripts under revision.

1. **Jangid, P.;** Chaudhury, S. *Anomalous diffusion of coupled active–passive particles in viscoelastic media*. Under peer review (2026).
2. **Jangid, P.;** Chaudhury, S.; Kolomeisky, A. *Dynamic Fluctuations Can Counteract Catalytic Inhibition and Enhance Catalytic Performance*. Under peer review (2026).
3. **Jangid, P.;** Chaudhury, S. *DNA target search in chromatin compartments under stochastic resetting*. *J. Phys. Chem. B* **2026**, 130, 3766–3775.
4. **Jangid, P.;** Chaudhury, S. *Stochastic dynamics of nanoparticle catalysis: A discrete-state perspective*. *Mater. Horiz.* **2026**, 13, 2152–2172.
5. **Jangid, P.;** Metzler, R.; Chaudhury, S. *Internal protein motion in a rough model potential*. *J. Chem. Phys.* **2025**, 163, 244101.
6. **Jangid, P.;** Chaudhury, S.; Kolomeisky, A. *Effect of temperature in dynamic catalysis*. *J. Phys. Chem. C* **2025**, 129, 15606–15618.
7. **Jangid, P.;** Chaudhury, S.; Kolomeisky, A. *Theoretical understanding of dynamic catalysis*. *J. Phys. Chem. C* **2024**, 128, 9077–9089.
8. **Jangid, P.;** Chaudhury, S. *Transition Path Dynamics of Non-Markovian Systems across a Rough Potential Barrier*. *J. Phys. Chem. A* **2024**, 128, 10041–10052.
9. **Jangid, P.;** Punia, B.; Chaudhury, S. *Stochastic dynamics of hairballs in single-polymer growth*.

Phys. Chem. Chem. Phys. **2024**, 26, 29749–29758.

10. Chaudhury, S.; **Jangid, P.**; Kolomeisky, A. B. *Dynamics of chemical reactions on single nanocatalysts with heterogeneous active sites. J. Chem. Phys.* **2023**, 158, 074101.
11. **Jangid, P.**; Chaudhury, S. *Exploring the role of heterogeneity in quorum sensing cells: A discrete-state stochastic approach. J. Stat. Mech.* **2023**, 2023, 093501.

AWARDS & FELLOWSHIPS

- 2026 ANRF-ITS Travel Grant, Anusandhan National Research Foundation (ANRF), India
- 2024 Best Poster Award, Royal Society of Chemistry – CrystEngComm Symposium, IISER Pune
- 2023 Best Teaching Assistant Award, Department of Chemistry, IISER Pune
- 2023 CSIR-UGC NET, Council of Scientific and Industrial Research (CSIR-UGC), India
- 2021 PhD Research Fellowship, IISER Pune
- 2021 Graduate Aptitude Test in Engineering (GATE), Indian Institute of Technology (IIT) Bombay, India
- 2020 Academic Excellence Award, IISER Mohali
- 2016 Joint Entrance Examination (JEE Advanced), Indian Institute of Technology (IIT) Guwahati, India

TEACHING

Worked as a Teaching Assistant (TA) in *Quantum Chemistry, Organic Chemistry, Thermodynamics* at IISER Pune, India.

PROFESSIONAL SERVICE

- 2025 Organizer, ChemSymphoria2025, IISER Pune. Managed conference finances and event logistics.